

Lecture 6: Some important large sample results
(Asymptotics)

I. Convergence in probability

Let $\{X_n\}_{n \geq 1}$ be a sequence of random variables defined on $(\Omega, \mathcal{F}, P)$

Then X_n is said to converge in probability to a random variable X ,

(notation) $X_n \xrightarrow{P} X$ as $n \rightarrow \infty$, if

$$P[|X_n - X| > \epsilon] \rightarrow 0 \text{ as } n \rightarrow \infty \quad \forall \epsilon > 0.$$

Eg., for the sequence $X_1, X_2, \dots, X_n, \dots$, define $Y_n = \sum_{i=1}^n X_i$, or $Z_n = \frac{1}{n} \sum_{i=1}^n X_i$.

Then ~~Y_n and Z_n~~ $\{Y_n\}_{n \geq 1}$ or $\{Z_n\}_{n \geq 1}$ are ~~seq.~~ ^{themselves} seq. of r.v.s

Now, if $\{X_n\}_{n \geq 1}$ is a sequence iid r.v.s with $E(X_i) = \theta$, say, then

$$Z_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \theta \text{ as } n \rightarrow \infty \text{ (directly from WLLN)}$$

If further, $V(X_i) = \sigma^2$, then we can apply the Chebyshev's Inequality

Chebyshev's Inequality Let X be a r.v. with finite mean and variance,
 $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$, respectively, then

$$P[|X - \mu|/\sigma \leq t] \geq 1 - \frac{1}{t^2} \quad \forall t > 0$$

$$\Rightarrow P[|X - \mu| > \epsilon] \leq \frac{\sigma^2}{\epsilon^2} \quad \forall \epsilon > 0$$

$$\text{so that, } P[|Z_n - \theta| > \epsilon] \leq \frac{\sigma^2}{n\epsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ \& } Z_n \xrightarrow{P} \theta$$

This is irrespective of any probability distribution.

Now, if $X_1, X_2, \dots$ are iid Ber(θ) (or, Bin(1, θ)) variables,

$$\text{then } P[|Z_n - \theta| > \epsilon] \leq \frac{\theta(1-\theta)}{n\epsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ so } Z_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \theta$$

$$\text{But } P[|Y_n - \theta| > \epsilon] \leq \frac{n\theta(1-\theta) + (n\theta - \theta)^2}{\epsilon^2} \not\rightarrow 0 \text{ as } n \rightarrow \infty, \text{ and } Y_n \not\xrightarrow{P} \theta$$

(Here, since $E(Y_n) \neq \theta$, $\text{Var}(Y_n)$ is replaced by $\text{MSE}(Y_n) = \text{Var}(Y_n) + \text{Bias}^2(Y_n)$)

Now (i) If $X_n \xrightarrow{P} X$, for a fixed c , $cX_n \xrightarrow{P} cX$

(ii) " " for a continuous fn g , $g(X_n) \xrightarrow{P} g(X)$

$$\text{(iii) If } X_n \xrightarrow{P} X, Y_n \xrightarrow{P} Y, \quad X_n \pm Y_n \xrightarrow{P} X \pm Y$$

$$X_n Y_n \xrightarrow{P} XY$$

$$X_n / Y_n \xrightarrow{P} X/Y,$$

provided $P(Y=0)=0$.

eg, $X_1, X_2, \dots, X_n, \dots$ iid $U(0, \theta)$, $\theta > 0$

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$$X_{(n)} \xrightarrow{P} \theta \quad (\text{check})$$

If true, then $e^{X_{(n)}} \xrightarrow{P} e^\theta$

$\sqrt{X_{(n)}} \xrightarrow{P} \sqrt{\theta}$, etc. are also true.

The Weak Law of Large Numbers (WLLN)

Let $\{X_n\}_{n \geq 1}$ be a seq. of r.v.s. $\{X_n\}$ satisfies the WLLN if $\exists$ sequence of constants $\{a_n\}, \{b_n\}$, $a_n \in \mathbb{R}$, $b_n > 0$, $b_n \uparrow \infty \Rightarrow$ if $S_n = \sum_{i=1}^n X_i$, then

$$\frac{S_n - a_n}{b_n} \xrightarrow{P} 0 \text{ as } n \rightarrow \infty.$$

Case (i) $X_1, X_2, \dots$ are iid variables with $E|X_i| < \infty$, then WLLN holds for $\{X_n\}_{n \geq 1}$ and $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} E(X_i)$. (sample mean convergence prob. to pop. mean)
(What are a_n & b_n here?)

(ii) Note (i) does not require existence of finite 2nd moments, ~~and~~
Now, if $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$, then similar convergence result for sample variance also holds true, i.e., in addition to $\bar{X}_n \rightarrow \mu$,

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \xrightarrow{P} \sigma^2. \text{ Also, } V_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \xrightarrow{P} \sigma^2.$$

(In the small sample sense, while S_n^2 is an unbiased estimator of σ^2 , V_n is a biased estimator) Both S_n^2 and V_n are consistent estimators of σ^2 (More later)

(iii) $X_1, X_2, \dots, X_n, \dots$ uncorrelated ~~having~~ $E(X_i) = \mu_i$, $\text{Var}(X_i) = \sigma_i^2$
(non iid-case), $\frac{S_n - \sum_{i=1}^n \mu_i}{n} \xrightarrow{P} 0$ as $n \rightarrow \infty$, provided $\frac{1}{n} \sum_{i=1}^n \sigma_i^2 \rightarrow 0$ as $n \rightarrow \infty$.

Under the ^{above} non-iid set-up, if $\sum \sigma_i^2 \rightarrow \infty$, then $\frac{S_n - \sum \mu_i}{\sum \sigma_i^2} \xrightarrow{P} 0$

$$\sigma_i^2 = \sigma^2 \forall i, \quad \frac{S_n - \sum \mu_i}{n \sigma^2} \xrightarrow{P} 0$$

II. Convergence in distribution/law

A sequence of r.v.s $\{X_n\}_{n \geq 1}$ convs in distribution/law to a r.v. X ($X_n \xrightarrow{L} X$) as $n \rightarrow \infty$, if $F_{X_n}(x) \rightarrow F_X(x)$ $\forall x$ at which F_X is continuous.
Note F_{X_n} and F_X are the cdf's of X_n and X , respectively.

eg 1) $x_1, x_2, \dots, x_n, \dots$ are iid $N(0,1)$ variables.

Then $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i = \bar{X}_n \sim N(0, \frac{1}{n})$. Check with Chebyshev's Ineq. $\bar{X}_n \xrightarrow{P} 0$.

Now, $F_{Z_n}(z) = P(\bar{X}_n \leq z) = \Phi(\sqrt{n} \bar{X}_n) \rightarrow \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$ as $n \rightarrow \infty$. (check!) which is the

cdf of Z , a degenerate r.v. at 0. with

$$F_Z(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$

Thus $Z_n \xrightarrow{L} 0$

2) $x_1, x_2, x_n, \dots$ are iid $U(0, \theta)$, $\theta > 0$.

$$Y_n = \max\{x_{(n)}\}$$

$$F_{Y_n}(y) = \begin{cases} 0 & \text{if } y < 0 \\ (y/\theta)^n & \text{if } 0 \leq y < \theta \\ 1 & \text{if } y \geq \theta \end{cases} \rightarrow F_Y(y) = \begin{cases} 0 & \text{if } y < \theta \\ 1 & \text{if } y \geq \theta \end{cases} \text{ as } n \rightarrow \infty$$

But F_Y is the c.d.f. of Y degenerate at θ ; and $Y_n = \max x_{(n)} \xrightarrow{L} \theta$.

Note = (i) If $X_n \xrightarrow{P} X$, then $X_n \xrightarrow{L} X$ as $n \rightarrow \infty$. The converse is not true, in general. However, if $X_n \xrightarrow{L} c$, fixed, then $X_n \xrightarrow{P} c$.

(ii) Slutsky's lemma. $X_n \xrightarrow{L} X$, $Y_n \xrightarrow{P} c$, fixed, then

$$X_n \pm Y_n \xrightarrow{L} X \pm c$$

$$X_n Y_n \xrightarrow{L} cX$$

$$X_n / Y_n \xrightarrow{L} X/c, \text{ provided } c \neq 0.$$

Central Limit Theorem

$x_1, x_2, \dots$ seq. of iid r.v.s with common cdf F .
 $\{A_n\}$ and $\{B_n\}$, $\exists A_n \in \mathbb{R}, B_n > 0$, and $S_n = \sum_{i=1}^n x_i$, then

$$\frac{S_n - A_n}{B_n} \xrightarrow{L} Z, \text{ which has a non-degenerate distn.}$$

($\Rightarrow P(\frac{S_n - A_n}{B_n} \leq z) \rightarrow G(z)$ at all continuity pts of G , the cdf. of Z .)

Case $\{x_n\}_{n \geq 1}$ seq. of iid r.v.s with $E(x_i) = \mu$, $\text{Var}(x_i) = \sigma^2$ if i, S_n

$$\text{then } \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} \xrightarrow{L} Z \sim N(0,1).$$

e.g., $x_1, x_2, \dots$ iid $\text{Exp}(\theta)$, then

$$\sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \theta^2).$$

$\rightarrow$ by CLT

The Δ -method $\{x_n\}_{n \geq 1}$ seq. of r.v.s $\Rightarrow \sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \sigma^2)$, for a real-valued differentiable fn g , $\sqrt{n}(g(\bar{X}_n) - g(\theta)) \xrightarrow{L} N(0, (g'(\theta))^2 \sigma^2)$, provided $g'(\theta) \neq 0$.
 e.g., $\text{at } \theta = 1$, $\sqrt{n}(\frac{1}{\bar{X}_n} - \frac{1}{\theta}) \xrightarrow{L} N(0, \frac{1}{\theta^2})$, by Δ -method.